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VENT ASSEMBLY FOR A SEALED ENCLOSURE

Abstract

A vent assembly for a sealed enclosure, such as a battery enclosure for an electric vehicle, is provided. The vent assembly includes a housing, a first valve, a second valve, and a protective cover. The housing includes a first valve mount and a second valve mount. The first valve is attached to the first valve mount, and the second valve is attached to the second valve mount. A bracket may also be included. The bracket includes a body and support arms extending from the body. The support arms are attached to an inner surface of the housing. The support arms are configured to attach the housing to an enclosure surface. The first valve is a unidirectional valve, while the second valve may be an equalization valve or a unidirectional valve.

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Background/Summary

INTRODUCTION

[0001] The present disclosure relates to electric vehicles (EVs). More particularly, the present disclosure relates to battery enclosures for electric vehicles.

SUMMARY

[0002] Embodiments of the present disclosure advantageously provide a vent assembly for a sealed enclosure, such as a battery enclosure for an electric vehicle.

[0003] In certain embodiments, a vent assembly includes a housing, a unidirectional valve, an equalization valve, and a protective cover. The housing includes a first valve mount and a second valve mount. The first valve mount defines a first valve opening and a first flow opening surrounding the first valve opening. The second valve mount defines a second valve opening. The unidirectional valve is attached to the first valve mount, and the equalization valve is attached to the second valve mount. In certain embodiments, the vent assembly may also include a bracket that has a body and support arms that extend from the body. The support arms are attached to a first surface of the housing, and are configured to attach the housing to an enclosure surface.

[0004] In certain embodiments, a vent assembly includes a housing, a first unidirectional valve, a second unidirectional valve, and a protective cover. The housing includes a first valve mount and a second valve mount. The first valve mount defines a first valve opening and a first flow opening surrounding the first valve opening. The second valve mount defines a second valve opening and a second flow opening surrounding the second valve opening. The first unidirectional valve is attached to the first valve mount, and the second unidirectional valve is attached to the second valve mount. In certain embodiments, the vent assembly may also include a bracket that has a body and support arms that extend from the body. The support arms are attached to a first surface of the housing, and are configured to attach the housing to an enclosure surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 depicts a diagram of an example electric vehicle, in accordance with embodiments of the present disclosure.

[0006] FIG. 2A presents an exploded perspective view of an example battery enclosure, in accordance with embodiments of the present disclosure.

[0007] FIGS. 2B, 2C present front and rear views of the example battery enclosure (respectively), in accordance with embodiments of the present disclosure.

[0008] FIGS. 3A, 3B present top and bottom views of an example vent assembly (respectively), in accordance with embodiments of the present disclosure.

[0009] FIGS. 3C, 3D present a top view and a close-up view of the example vent assembly (respectively), in accordance with embodiments of the present disclosure.

[0010] FIGS. 3E, 3F present cross-sectional views of an example unidirectional valve in a closed position and an open position (respectively), in accordance with embodiments of the present disclosure.

[0011] FIGS. 4A, 4B present top and bottom views of another example vent assembly (respectively), in accordance with embodiments of the present disclosure.

[0012] FIG. 4C presents a bottom view of the other example vent assembly without a bracket and adhesive, in accordance with embodiments of the present disclosure.

[0013] FIGS. 4D, 4E present a top view and a close-up view of the other example vent assembly (respectively), in accordance with embodiments of the present disclosure.

[0014] FIGS. 4F, 4G present two perspective views of an example equalization valve, in accordance with embodiments of the present disclosure.

[0015] FIGS. 5A, 5B present two top views of example openings for mounting vent assemblies

(respectively) in a battery enclosure, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0016] This disclosure is generally directed to a vent assembly for a sealed enclosure, such as a battery enclosure for an electric vehicle.

[0017] A battery enclosure for an electric vehicle provides a sealed housing to protect the battery pack from exposure to water, dust, debris and other elements. In one example, the battery enclosure may include, inter alia, a battery pack, a bottom tray that supports the battery pack, a frame attached to the bottom tray, and a top cover (or tub) attached to the frame (and/or the bottom tray). In another example, the battery enclosure may include a battery pack, a bottom tub that supports the battery pack, a frame attached to the bottom tub, and a top cover (or tray) attached to the frame (and/or the bottom tub).

[0018] The top cover may be sealed to the bottom tray (or tub), and the frame may include one or more longitudinal members that are located within the enclosed space formed by the top cover and the bottom tray or tub. Alternatively, the frame may include transverse and longitudinal members that form a rectangular outer frame, the top cover may be sealed to the upper surface of the frame, and the bottom tray or tub may be sealed to the lower surface of the frame. Additionally, one or more cooling plates may be attached to, or integrally formed with, the bottom tray (or tub) and/or the top cover. Alternatively, one or more cooling plates may be attached to the sides of the battery pack, located within the battery pack, etc.

[0019] In a further example, the battery enclosure may include a battery pack, a top tub that supports the battery pack, a frame attached to the top tub, and a bottom cover (or tray) attached to the frame (and/or the top tub). Other configurations of the battery enclosure are also possible.

[0020] A battery enclosure for an electric vehicle has strict structural and sealing requirements. The structural requirements may include, inter alia, a satisfactory reaction to an impact load, such as an underside impact (e.g., an impact to the underside of the vehicle), a side pole impact, a frontal offset impact, etc. The sealing requirements may include, inter alia, protection against water ingress, remaining sealed through 5 minutes of thermal runaway, etc.

[0021] Importantly, the battery enclosure not only equalizes the air pressure within the enclosed space to accommodate gradual changes in atmospheric pressure outside the enclosed space, but also vents overpressure conditions caused by thermal runaway events to the atmosphere.

[0022] Embodiments of the present disclosure advantageously provide a vent assembly for a sealed enclosure, such as a battery enclosure for an electric vehicle. The vent assembly includes a housing, a first valve, a second valve, and a bracket. The housing includes a first valve mount and a second valve mount. The first valve mount defines a first valve opening and a first flow opening that surrounds the first valve opening. The second valve mount defines at least a second valve opening. The first valve is a unidirectional valve. In certain embodiments, the second valve may be an equalization valve. In other embodiments, the second valve may be a unidirectional valve, and the second valve mount may also include a second flow opening that surrounds the second valve opening. The bracket includes a body and support arms that extend from the body. The support arms are attached to a first surface of the housing (such as an inner surface of the housing), and are configured to attach the housing to an enclosure surface.

[0023] An equalization valve (such as a breather valve, a breather membrane, etc.) does not need a flow opening, and not only equalizes the air pressure within the sealed enclosure, but also prevents moisture from being admitted to the sealed enclosure. A unidirectional valve (such as an umbrella valve, etc.) vents overpressure conditions within the sealed enclosure to the atmosphere and prevents oxygen from being admitted to the sealed enclosure.

[0024] In certain embodiments, a protective cover may be attached to a second surface of the housing (such as an outer surface of the housing) to protect the first and second valves from debris, etc. The protective cover may include at least one rectangular slit that begins on the perimeter (outer edge) of the protective cover and extends inward a relatively short distance (such as 5 mm,

10 mm, 15 mm, etc.). The rectangular slit(s) mitigate the transmission of shock and vibration to the protective cover to avoid damage, such as tears, adhesive releases, etc. The rectangular slit may also promote the opening, tearing, etc., of the protective cover to allow the unidirectional valve to open during an overpressure event, such as thermal runaway, etc. Other slit shapes and locations are also supported, such as a “+” shape or an “x” shape disposed in the center of the protective cover, etc. The protective cover may be plastic film with adhesive applied along the perimeter, double-sided tape, etc.

[0025] FIG. 1 depicts a diagram of electric vehicle **100**, in accordance with embodiments of the present disclosure.

[0026] Electric vehicle **100** includes, inter alia, a frame and body **110**, an electrical power storage and distribution system, a propulsion system, a suspension system, a steering system, auxiliary and accessory systems (such as thermal management, lighting, wireless communications, navigation, etc.), etc.

[0027] Generally, body **110** may be directly or indirectly mounted to a frame (i.e., body-on-frame construction), or body **110** may be formed integrally with a frame (i.e., unibody construction). Body **110** includes, inter alia, front end **120**, front light bar **122**, front turn lights **123**, stadium light ring **124**, headlights **126**, charging port **130** with charging port cover **136** concealing charging connector socket, driver/passenger compartment or cabin **140**, bed **150**, rear end **160** with rear tail lights **162**, a rear light bar, etc. Electric vehicle **100** may be a pickup truck, a sport utility vehicle (SUV) in which bed **150** is replaced by an extension of cabin **140**, or a sedan in which bed **150** is replaced by a trunk. In certain embodiments, electric vehicle may be an electric delivery vehicle, an electric cargo van, etc.

[0028] The propulsion system may include, inter alia, one or more electronic control units (ECUs), one or more electronic drive units (EDUs), wheels **170**, etc. The electrical power storage and distribution system may include, inter alia, one or more ECUs, a battery enclosure containing a battery pack including one or more batteries or battery modules, a vehicle charging subsystem including charging port **130**, high voltage (HV) cables, etc.

[0029] FIG. 2A presents an exploded perspective view of battery enclosure **200**, in accordance with embodiments of the present disclosure.

[0030] In certain embodiments, battery enclosure **200** includes, inter alia, top cover **210**, bottom tray **220**, frame members **230.1**, **230.2**, batteries **240.1**, **240.2**, **240.3**, lower plate **250**, and electronics enclosure **260**. Top cover **210** is attached to bottom tray **220**, frame members **230.1**, **230.2** are attached to top cover **210** and bottom tray **220**, batteries **240.1**, **240.2**, **240.3** are attached to bottom tray **220**, and lower plate **250** is attached to bottom tray **220** using fasteners, such as screws, bolts and nuts, rivets, snap-fit clips, pins, etc. Battery enclosure **200** has a generally rectangular shape, with a front side, a left side, a right side, and a rear side.

[0031] Top cover **210** may be a cold stamped steel deep drawn cover, while bottom tray **220** may be a hot stamped steel tray. A sealing material may be disposed between top cover **210** and bottom tray **220** to provide a continuous seal along the periphery of battery enclosure **200**. The sealing material may protect against water intrusion, thermal runaway, etc., as noted above. In certain embodiments, three vent assemblies **300** may be mounted to the front side of top cover **210** (also depicted in FIG. 2B), and three vent assemblies **400** may be mounted to the rear side of top cover **210** (depicted in FIG. 2C). In other embodiments, other numbers of vent assemblies **300**, **400** and mounting locations can be implemented.

[0032] A cooling plate may be attached to, or integrally formed with, bottom tray **220**. Coolant manifold **222** may be attached to bottom tray **220** and coupled to the cooling plate, while coolant inlet couplings **224** may be attached to top cover **210** and coupled to coolant manifold **222**.

[0033] Frame members **230.1**, **230.2** may be longitudinal steel members that are located within the enclosed space formed by top cover **210** and bottom tray **220**.

[0034] Batteries **240.1**, **240.2**, **240.3** form the battery pack, and each battery **240.1**, **240.2**, **240.3**

may include a number of battery cells, such as cylindrical cells, prismatic cells, pouch cells, etc. In certain embodiments, each battery **240.1**, **240.2**, **240.3** may be a battery module.

[0035] Lower plate **250** and electronics enclosure **260** may be formed from a composite material. Generally, lower plate **250** protects bottom tray **220** from road or road debris damage.

[0036] Electronics enclosure **260** may be mounted on top cover **210**, and may contain certain electrical power system components, such as an energy management module (EMM), a high voltage distribution box (HVDB), etc., coupled to batteries **240.1**, **240.2**, **240.3**.

[0037] FIGS. **2B**, **2C** present front and rear views of battery enclosure **200** (respectively), in accordance with embodiments of the present disclosure.

[0038] As depicted in FIG. **2B**, vent assemblies **300** and coolant inlet couplings **224** are mounted to the front side of top cover **210**. Lower plate **250** and electronics enclosure **260** are also visible. While three vent assembly **300** are depicted, a different number of vent assemblies **300** may be attached to the front side of top cover **210**, such as one, two, four, etc. Additionally, vent assemblies **400** (depicted in FIG. **2C**) may be mounted to the front side of top cover **210**, or a combination of vent assemblies **300** and vent assemblies **400** may be mounted to the front side of top cover **210**.

[0039] As depicted in FIG. **2C**, vent assemblies **400** are mounted to the rear side of top cover **210**. A portion of bottom tray **220**, lower plate **250**, and electronics enclosure **260** are also visible. While three vent assembly **400** are depicted, a different number of vent assemblies **400** may be attached to the rear side of top cover **210**, such as one, two, four, etc. Additionally, vent assemblies **300** (depicted in FIG. **2B**) may be mounted to the rear side of top cover **210**, or a combination of vent assemblies **300** and vent assemblies **400** may be mounted to the rear side of top cover **210**.

[0040] FIGS. **3A**, **3B** present top and bottom views of vent assembly **300** (respectively), in accordance with embodiments of the present disclosure.

[0041] FIG. **3A** presents a top view of vent assembly **300**, in which the external surfaces and components of vent assembly **300** are depicted. When vent assembly **300** is mounted and sealed to battery enclosure **200**, the external surfaces and components of vent assembly **300** are exposed to the atmosphere.

[0042] Vent assembly **300** includes, inter alia, housing **310**, outer portion **320**, inner portion **330**, bracket **340** (FIG. **3B**), unidirectional valve **360**, and unidirectional valve **362**.

[0043] Housing **310** may be mild steel (such as nickel-coated steel, zinc-coated steel, etc.), aluminum, etc. Housing **310** may be formed by a cold-stamping process, a hot-stamping process, etc. Similarly, bracket **340** may be mild steel, aluminum, etc., and may be formed by a cold-stamping process, a hot-stamping process, etc. Bracket **340** may be attached to housing **310** by welding, adhesive bond, etc. In certain embodiments, housing **310** and bracket **340** may be formed from plastic, such as injection-molded plastic, etc., and may be attached using heat staking or other assembly joining processes.

[0044] FIG. **3B** presents a bottom view of vent assembly **300**, in which the internal surfaces and components of vent assembly **300** are depicted. When vent assembly **300** is mounted and sealed to battery enclosure **200**, certain internal surfaces and components of vent assembly **300** are exposed to the enclosed space (or interior) of battery enclosure **200**, while other surfaces and components contact the external surface of battery enclosure **200**.

[0045] In certain embodiments, adhesive **322** may be applied to the bottom surface of outer portion **320**. When vent assembly **300** is mounted to battery enclosure **200**, adhesive **322** secures and seals housing **310** to the external surface of battery enclosure **200**. In other embodiments, a sealing material may be applied to the bottom surface of outer portion **320** instead of adhesive **322**. When vent assembly **300** is mounted to battery enclosure **200**, the sealing material seals housing **310** to battery enclosure **200**, while bracket **340** secures housing **310** to battery enclosure **200**, as described below.

[0046] Adhesive **322** may be a urethane adhesive, an acrylic adhesive, double-sided, pressure sensitive adhesive (PSA) tape, 2-part epoxy, etc. Generally, adhesive **322** may have a width of 4

mm, 7 mm, 10 mm, etc., and a thickness of 1.5 mm, 2 mm, 2.5 mm, 3.5 mm, etc. In certain embodiments, adhesive **322** may have a coverage area of between about 1,500 mm² and about 3,400 mm², and may withstand a pressure of about 50 kPa, which translates to a tensile stress of about 10 psi to about 20 psi.

[0047] Inner portion **330** extends from outer portion **320**, and includes valve mount **350** and valve mount **352**. In FIG. 3A, inner portion **330** is recessed with respect to outer portion **320**, while in FIG. 3B, inner portion **330** is raised with respect to outer portion **320**.

[0048] Bracket **340** includes body **340b** and support arms **341**. In certain embodiments, body **340b** may contact inner portion **330**, while in other embodiments, body **340b** may be separated from inner portion **330** by a small air gap, such as 1 mm, 3 mm, 5 mm, etc. Support arms **341** extend from body **340b** and attach to outer portion **320**. Advantageously, support arms **341** are configured to be received by mounting recesses when vent assembly **300** is mounted in an opening in battery enclosure **200**, such as recesses **241** that are adjacent to opening **242** (as depicted in FIG. 5A).

Generally, support arms **341** cooperate with recesses **241** to align vent assembly **300** in opening **242**. In certain embodiments, support arms **341** may also provide a predetermined clamping force.

[0049] In certain embodiments, support arms **341** align vent assembly **300** within opening **242**, and adhesive **322** secures vent assembly **300** to battery enclosure **200**. In certain embodiments, support arms **341** align vent assembly **300** within opening **242**, and both support arms **341** and adhesive **322** secure vent assembly **300** to battery enclosure **200**. In certain embodiments, support arms **341** aligns and secures vent assembly **300** to battery enclosure **200**, while a sealing material seals vent assembly **300** to battery enclosure **200**.

[0050] Unidirectional valve **360** is mounted to valve mount **350**, and unidirectional valve **362** is mounted to valve mount **352**. Referring to FIG. 3E, unidirectional valves **360**, **362** may be umbrella valves that include valve disc **364**, valve stem **365** that extends from valve disc **364**, and end portion **366** that is disposed on the end of valve stem **365**. End portion **366** has a truncated cone shape, and is configured to engage valve mount **350** and valve mount **352**. Generally, unidirectional valves **360**, **362** are made from a deformable material, such as elastomer rubber, silicone, fluorosilicone, liquid silicone rubber (LSR), etc.

[0051] Referring back to FIG. 3B with respect to valve mount **350**, inner portion **330** defines valve opening **332** and flow opening **334** that surrounds valve opening **332**. Valve stem **365** of unidirectional valve **360** is disposed within valve opening **332**, and end portion **366** of unidirectional valve **360** engages the inner surface of inner portion **330** (FIGS. 3E, 3F). In certain embodiments, flow opening **334** includes a number of flow opening segments **333**, such as two segments, three segments, four segments, five segments (depicted in FIG. 3B), six segments, etc. Flow opening segments **333** are separated by a number of flow opening arms **335** (portions visible), such as two arms, three arms, four arms, five arms (depicted in FIG. 3B), six arms, etc.

[0052] With respect to valve mount **350**, bracket **340** defines valve passage **342** and flow opening **344** that surrounds valve passage **342**. In certain embodiments, flow opening **344** includes a number of flow opening segments **343**, such as two segments, three segments, four segments, five segments (depicted in FIG. 3B), six segments, etc. Flow opening segments **343** are separated by a number of flow opening arms **345**, such as two arms, three arms, four arms, five arms (depicted in FIG. 3B), six arms, etc. Generally, flow opening arms **345** align with flow opening arms **335**, valve passage **342** is larger than valve opening **332**, and flow opening **344** is larger than flow opening **334**.

[0053] Similarly, with respect to valve mount **352**, inner portion **330** defines valve opening **336** and flow opening **338** that surrounds valve opening **336**. Valve stem **365** of unidirectional valve **362** is disposed within valve opening **336**, and end portion **366** of unidirectional valve **360** engages the inner surface of inner portion **330** (FIGS. 3E, 3F). In certain embodiments, flow opening **338** includes a number of flow opening segments **337**, such as two segments, three segments, four segments, five segments (depicted in FIG. 3B), six segments, etc. Flow opening segments **337** are

separated by a number of flow opening arms **339** (portions visible), such as two arms, three arms, four arms, five arms (depicted in FIG. **3B**), six arms, etc.

[0054] With respect to valve mount **352**, bracket **340** defines valve passage **346** and flow opening **348** that surrounds valve passage **346**. In certain embodiments, flow opening **348** includes a number of flow opening segments **347**, such as two segments, three segments, four segments, five segments (depicted in FIG. **3B**), six segments, etc. Flow opening segments **347** are separated by a number of flow opening arms **349**, such as two arms, three arms, four arms, five arms (depicted in FIG. **3B**), six arms, etc. Generally, flow opening arms **349** align with flow opening arms **339**, valve passage **346** is larger than valve opening **336**, and flow opening **348** is larger than flow opening **338**.

[0055] FIGS. **3C**, **3D** present a top view and a close-up view of vent assembly **302** (respectively), in accordance with embodiments of the present disclosure.

[0056] Vent assembly **302** includes all of the elements of vent assembly **300** described above with respect to FIGS. **3A**, **3B**. Housing **310** and outer portion **320** are identified.

[0057] In certain embodiments, vent assembly **302** may include protective cover **380** disposed over unidirectional valves **360**, **362**, inner portion **330**, and about half of outer portion **320**. Protective cover **380** protects unidirectional valves **360**, **362** from debris, etc. Generally, protective cover **380** may be secured to outer portion **320** using adhesive. For example, protective cover **380** may be a plastic film with adhesive along the perimeter, double-sided tape, etc. The thickness of protective cover may be about 0.3 mm, 0.5 mm, 0.7 mm, etc.

[0058] In certain embodiments, protective cover **380** may include at least one rectangular slit **382** that begins on the perimeter (outer edge) of protective cover **380** and extends inward toward unidirectional valves **360**, **362** a relatively short distance (such as 5 mm, 10 mm, 15 mm, etc.). Rectangular slit **382** may mitigate the transmission of shock and vibration to protective cover **380** to avoid damage, such as tears, adhesive releases, etc. Other slit shapes and locations are also supported, such as a "+" shape or an "x" shape disposed in the center of protective cover **380**, etc.

[0059] In certain embodiments, outer portion **320** may include two or more bosses **324** that extend upward from the surface of outer portion **320**. Each boss defines an inner cavity in which one support arm **341** of bracket **340** may be secured. Advantageously, boss **324** is not located in the coverage area of adhesive **322**, and the support arm **341** that is secured within boss **324** may have an extended length to provide additional clamping force.

[0060] FIGS. **3E**, **3F** present cross-sectional views of unidirectional valve **360** in a closed position and an open position (respectively), in accordance with embodiments of the present disclosure.

[0061] As discussed above, unidirectional valve **360** may be an umbrella valve that includes valve disc **364**, valve stem **365** that extends from valve disc **364**, and end portion **366** that is disposed on the end of valve stem **365**. End portion **366** has a truncated cone shape. Valve stem **365** is disposed within valve opening **332**, and end portion **366** engages the surface of inner portion **330**.

[0062] FIG. **3E** depicts unidirectional valve **360** in the closed position. Valve disc **364** is configured to form a seal with the surface of inner portion **330** of housing **310** to prevent airflow through flow opening segments **333** of inner portion **330** and flow opening segments **343** of bracket **340**.

[0063] FIG. **3F** depicts unidirectional valve **360** in the open position. Valve disc **364** is configured to extend above the surface of outer portion **320** of housing **310** to allow airflow through flow opening segments **333** of inner portion **330** and flow opening segments **343** of bracket **340**. The transition from the closed position to the open position is based on the pressure difference between the enclosed space of battery enclosure **200** and the atmosphere. In certain embodiments, the pressure difference may be 1.5 to 30 kPa.

[0064] With respect to protective cover **380** described above, when unidirectional valve **360** is in the open position, protective cover **380** does not prevent airflow through flow opening segments **333** of inner portion **330** and flow opening segments **343** of bracket **340**.

[0065] FIGS. **4A**, **4B** present top and bottom views of vent assembly **400** (respectively), in

accordance with embodiments of the present disclosure.

[0066] FIG. 4A presents a top view of vent assembly **400**, in which the external surfaces and components of vent assembly **400** are depicted. When vent assembly **400** is mounted and sealed to battery enclosure **200**, the external surfaces and components of vent assembly **400** are exposed to the atmosphere.

[0067] Vent assembly **400** includes, inter alia, housing **410**, outer portion **420**, inner portion **430**, bracket **440** (FIG. 4B), unidirectional valve **360**, and equalization valve **460**.

[0068] Similar to housing **310**, housing **410** may be mild steel (such as nickel-coated steel, zinc-coated steel, etc.), aluminum, etc., and may be formed by cold-stamping process, a hot-stamping process, etc. Similar to bracket **340**, bracket **440** may be formed from mild steel, aluminum, etc., and may be formed by cold-stamping process, a hot-stamping process, etc.

[0069] FIG. 4B presents a bottom view of vent assembly **400**, in which the internal surfaces and components of vent assembly **400** are depicted, in accordance with embodiments of the present disclosure. When vent assembly **400** is mounted and sealed to battery enclosure **200**, certain internal surfaces and components of vent assembly **400** are exposed to the enclosed space (or interior) of battery enclosure **200**, while other surfaces and components contact the external surface of battery enclosure **200**.

[0070] In certain embodiments, adhesive **422** may be applied to the bottom surface of outer portion **420**. When vent assembly **400** is mounted to battery enclosure **200**, adhesive **422** secures and seals housing **410** to battery enclosure **200**. In other embodiments, a sealing material may be applied to the inner surface of outer portion **420** instead of adhesive **422**. When vent assembly **400** is mounted to battery enclosure **200**, the sealing material seals housing **410** to battery enclosure **200**, while bracket **440** secures housing **410** to battery enclosure **200**, as described below. Adhesive **422** is the same as adhesive **322** described above.

[0071] Inner portion **430** extends from outer portion **420**, and includes valve mount **450** and valve mount **452**. In FIG. 4A, inner portion **430** is recessed with respect to outer portion **420**, while in FIG. 4B, inner portion **430** is raised with respect to outer portion **420**.

[0072] Bracket **440** includes body **440b** and support arms **441**. In certain embodiments, body **440b** may contact inner portion **430**, while in other embodiments, body **440b** may be separated from inner portion **430** by a small air gap, such as 1 mm, 3 mm, 5 mm, etc. Support arms **441** extend from body **440b** and attach to outer portion **420**. Advantageously, support arms **441** are configured to be received by mounting recesses when vent assembly **400** is mounted in an opening in battery enclosure **200**, such as recesses **241** adjacent to opening **244** (as depicted in FIG. 5B). Generally, support arms **441** cooperate with recesses **241** to align vent assembly **400** in opening **244**. In certain embodiments, support arms **441** may also provide a predetermined clamping force.

[0073] In certain embodiments, support arms **441** align vent assembly **400** within opening **244**, and adhesive **422** secures vent assembly **400** to battery enclosure **200**. In certain embodiments, support arms **441** align vent assembly **400** within opening **244**, and both support arms **441** and adhesive **422** secure vent assembly **400** to battery enclosure **200**. In certain embodiments, support arms **441** aligns and secures vent assembly **400** to battery enclosure **200**, while a sealing material seals vent assembly **400** to battery enclosure **200**.

[0074] Unidirectional valve **360** is mounted to valve mount **450**, and equalization valve **460** mounted to valve mount **452**. Equalization valve **460** may be a breather valve, a breather membrane, etc. Referring to FIGS. 4E, 4F, equalization valve **460** include valve membrane **467**, valve stem **465** that extends from valve membrane **467**, and end portion **466** that is disposed on the end of valve stem **465**. End portion **466** has a segmented cone shape, and is configured to engage valve mount **452**.

[0075] Referring back to FIG. 4B with respect to valve mount **450**, inner portion **430** defines valve opening **432** and flow opening **434** that surrounds valve opening **432**. Valve stem **365** of unidirectional valve **360** is disposed within valve opening **432**, and end portion **366** of

unidirectional valve **360** engages the inner surface of inner portion **430**. In certain embodiments, flow opening **434** includes a number of flow opening segments **433**, such as two segments, three segments, four segments, five segments (depicted in FIG. **4B**), six segments, etc. Flow opening segments **433** are separated by a number of flow opening arms **435** (portions visible), such as two arms, three arms, four arms, five arms (depicted in FIG. **4B**), six arms, etc.

[0076] With respect to valve mount **450**, bracket **440** defines valve passage **442** and flow opening **444** that surrounds valve passage **442**. In certain embodiments, flow opening **444** includes a number of flow opening segments **443**, such as two segments, three segments, four segments, five segments (depicted in FIG. **4B**), six segments, etc. Flow opening segments **443** are separated by a number of flow opening arms **445**, such as two arms, three arms, four arms, five arms (depicted in FIG. **4B**), six arms, etc. Generally, flow opening arms **445** align with flow opening arms **435**, valve passage **442** is larger than valve opening **432**, and flow opening **444** is larger than flow opening **434**.

[0077] With respect to valve mount **452**, outer portion **420** defines valve opening **446**. Valve stem **465** of equalization valve **460** is disposed within valve opening **446**, and end portion **466** of equalization valve **460** engages the inner surface of outer portion **420**.

[0078] FIG. **4C** presents a bottom view of vent assembly **400** without bracket **440** and adhesive **422**, in accordance with embodiments of the present disclosure.

[0079] Housing **410**, outer portion **420**, inner portion **430**, valve mount **450**, and valve mount **452** are identified. With respect to unidirectional valve **360**, valve disc **364** and end portion **366** are identified. With respect to equalization valve **460**, end portion **466** is identified.

[0080] With respect to valve mount **450**, valve opening **432**, flow opening **434**, flow opening segments **433** (five segments are depicted in FIG. **4C**), and flow opening arms **435** (five arms are depicted in FIG. **4C**) are identified.

[0081] With respect to valve mount **452**, end portion **466** is identified.

[0082] FIGS. **4D**, **4E** present a top view and a close-up view of vent assembly **402** (respectively), in accordance with embodiments of the present disclosure.

[0083] Vent assembly **402** includes all of the elements of vent assembly **400** described above with respect to FIGS. **4A**, **4B**, **4C**. Housing **410** and outer portion **420** are identified.

[0084] In certain embodiments, vent assembly **402** may include protective cover **480** disposed over unidirectional valve **360**, inner portion **430**, and a part of outer portion **420**. Protective cover **480** protects unidirectional valve **360** from debris, etc. Generally, protective cover **480** may be secured to outer portion **420** using adhesive. For example, protective cover **480** may be a plastic film with adhesive along the perimeter, double-sided tape, etc. The thickness of protective cover may be about 0.1 mm to about 2 mm, such as 0.3 mm, 0.5 mm, 0.7 mm, etc.

[0085] In certain embodiments, protective cover **480** may include at least one rectangular slit **482** that begins on the perimeter (outer edge) of protective cover **480** and extends inward toward unidirectional valve **360** a relatively short distance (such as 5 mm, 10 mm, 15 mm, etc.).

Rectangular slit **482** may mitigate the transmission of shock and vibration to protective cover **480** to avoid damage, such as tears, adhesive releases, etc. Other slit shapes and locations along the periphery are also supported, such as a “+” shape or an “x” shape disposed in the center of protective cover **480**, etc.

[0086] FIGS. **4F**, **4G** present two perspective views of equalization valve **460**, in accordance with embodiments of the present disclosure.

[0087] As discussed above, equalization valve **460** includes valve membrane **467**, valve stem **465** that extends from valve membrane **467**, and end portion **466** that is disposed on the end of valve stem **465**. End portion **466** has a segmented cone shape, and is configured to engage valve mount **452**.

[0088] FIGS. **5A**, **5B** present two top views of openings **242**, **244** for mounting vent assemblies **300**, **400** (respectively) in battery enclosure **200**, in accordance with embodiments of the present

disclosure.

[0089] Opening **242** is generally the same shape as housing **310** but smaller overall, and includes a recess **241** for each support arm **341** of vent assembly **300, 302**.

[0090] Similarly, opening **244** is generally the same shape as housing **410** but smaller overall, and includes a recess **241** for each support arm **441** of vent assembly **400, 402**.

[0091] Generally, the features described above for vent assemblies **300, 302, 400, 402** provide many advantages. Fasteners, such as bolts, etc., are not required to secure vent assemblies **300, 302, 400, 402** to battery enclosure **200**. Vent assemblies **300, 302, 400, 402** are easy to assemble, repair, and rework from the outside of battery enclosure **200**. Unidirectional valves **360, 362** minimize O.sub.2 intrusion during thermal propagation and venting. Vent assemblies **300, 302, 400, 402** have a minimal form factor, and are generally agnostic to the surface flatness, surface roughness, manufacturing process (such as extrusion, casting, stamping, composite, etc.) of battery enclosure **200**. Vent assemblies **300, 302, 400, 402** may be installed using a wedge to direct the gas outflow in a particular direction, such as up, down, or to the side of battery enclosure **200**. Flow opening arms **345, 349** of bracket **340** may be shaped to redirect gas outflow through flow opening segments **333** in a particular direction, such as up, down, or to the side of battery enclosure **200**.

[0092] The many features and advantages of the disclosure are apparent from the detailed specification, and, thus, it is intended by the appended claims to cover all such features and advantages of the disclosure which fall within the scope of the disclosure. Further, since numerous modifications and variations will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation illustrated and described, and, accordingly, all suitable modifications and equivalents may be resorted to that fall within the scope of the disclosure.

Claims

1. A vent assembly, comprising: a housing comprising: a first valve mount defining a first valve opening and a first flow opening surrounding the first valve opening, and a second valve mount defining a second valve opening; a unidirectional valve attached to the first valve mount; an equalization valve attached to the second valve mount; and a protective cover attached to a first surface of the housing above the valve disc, wherein, when the unidirectional valve is disposed in the open position, the protective cover does not prevent airflow through the first flow opening.
2. The vent assembly of claim 1, wherein: the unidirectional valve includes a valve disc and a valve stem extending from the valve disc; the valve stem is disposed within the first valve opening; and the valve stem includes an end portion that engages the first valve mount.
3. The vent assembly of claim 2, wherein: when the unidirectional valve is disposed in a closed position, the valve disc is configured to form a seal with the first surface of the housing to prevent airflow through the first flow opening; and when the unidirectional valve is disposed in an open position, the valve disc is configured to extend above the first surface of the housing to allow airflow through the first flow opening.
4. The vent assembly of claim 3, wherein the first flow opening includes a plurality of flow opening segments.
5. The vent assembly of claim 3, further comprising: a bracket comprising: a body, and support arms extending from the body and attached to a second surface of the housing, wherein the support arms are configured to align the housing to an enclosure surface.
6. The vent assembly of claim 5, wherein the body of the bracket defines: a first valve passage that is larger than the first valve opening of the first valve mount; a first flow opening that is larger than the first flow opening of the first valve mount; and a second valve passage that is larger than the second valve opening.
7. The vent assembly of claim 1, wherein the support arms are configured to provide a

predetermined clamping force.

8. The vent assembly of claim 7, further comprising adhesive disposed on the second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.

9. The vent assembly of claim 1, further comprising adhesive disposed on a second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.

10. A vent assembly, comprising: a housing comprising: a first valve mount defining a first valve opening and a first flow opening surrounding the first valve opening, and a second valve mount defining a second valve opening and a second flow opening surrounding the second valve opening; a first unidirectional valve attached to the first valve mount; a second unidirectional valve attached to the second valve mount; and a protective cover attached to a first surface of the housing above the first and second unidirectional valves, wherein, when the first unidirectional valve is disposed in the open position, the protective cover does not prevent airflow through the first flow opening; and wherein, when the second unidirectional valve is disposed in the open position, the protective cover does not prevent airflow through the second flow opening.

11. The vent assembly of claim 10, wherein: each unidirectional valve includes a valve disc and a valve stem extending from the valve disc, the valve stem including an end portion; the valve stem for the first unidirectional valve is disposed within the first valve opening; the valve stem for the second unidirectional valve is disposed within the second valve opening; the end portion of the first unidirectional valve engages the first valve mount; and the end portion of the second unidirectional valve engages the second valve mount.

12. The vent assembly of claim 11, wherein: when the first unidirectional valve is disposed in a closed position, the valve disc is configured to form a seal with the first surface of the housing to prevent airflow through the first flow opening; when the first unidirectional valve is disposed in an open position, the valve disc is configured to extend above the first surface of the housing to allow airflow through the first flow opening; when the second unidirectional valve is disposed in the closed position, the valve disc is configured to form a seal with a first surface of the housing to prevent airflow through the second flow opening; and when the second unidirectional valve is disposed in the open position, the valve disc is configured to extend above the surface of the housing to allow airflow through the second flow opening.

13. The vent assembly of claim 12, wherein the first flow opening includes a plurality of first flow opening segments, and the second flow opening includes a plurality of second flow opening segments.

14. The vent assembly of claim 10, further comprising: a bracket comprising: a body, and support arms extending from the body and attached to a second surface of the housing, wherein the support arms are configured to align the housing to an enclosure surface.

15. The vent assembly of claim 14, wherein the body of the bracket defines: a first valve passage that is larger than the first valve opening of the first valve mount; a first flow opening that is larger than the first flow opening of the first valve mount; and a second valve passage that is larger than the second valve opening of the second valve mount; and a second flow opening that is larger than the second flow opening of the second valve mount.

16. The vent assembly of claim 14, wherein the support arms are configured to provide a predetermined clamping force.

17. The vent assembly of claim 16, further comprising adhesive disposed on the second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.

18. The vent assembly of claim 10, further comprising adhesive disposed on a second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.

19. A method, comprising: providing a vent assembly for a sealed enclosure, the vent assembly including: a housing comprising: a first valve mount defining a first valve opening and a first flow opening surrounding the first valve opening, a second valve mount defining at least a second valve opening; a first valve attached to the first valve mount; a second valve attached to the second valve

mount; and a protective cover attached to a first surface of the housing above the valve disc, wherein, when the unidirectional valve is disposed in the open position, the protective cover does not prevent airflow through the first flow opening. wherein: the first valve is a unidirectional valve, the second valve is an equalization valve or a unidirectional valve, and the support arms are configured to align the housing to an enclosure surface.

20. The method of claim 19, wherein the second valve is the equalization valve.

21. The method of claim 19, wherein the second valve is the unidirectional valve; the second valve mount further includes a second flow opening that surrounds the second valve opening; the vent assembly further includes a protective cover attached to the first surface of the housing above the first valve and the second valve; and when the unidirectional valve is disposed in the open position, the protective cover does not prevent airflow through the first flow opening.

22. The method of claim 19, wherein the vent assembly further includes: a bracket comprising: a body, and support arms extending from the body and attached to a second surface of the housing, wherein the support arms are configured to align the housing to an enclosure surface.

23. The method of claim 22, wherein the vent assembly further includes adhesive disposed on the second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.

24. The method of claim 19, wherein the vent assembly further includes adhesive disposed on a second surface of the housing, the adhesive configured to attach and seal the housing to the enclosure surface.
